Human-in-the-Loop Formation-Containment Safe Control for Multi-agent Systems via Reinforcement Learning

Luning Yang, Pei Chi, Jiang Zhao, and Yingxun Wang

Abstract—**This article designs the optimal formation containment safe tracking control protocol for nonlinear second-order multi-agent systems (MASs) with unknown dynamics. A human operator can interact with the tracking leader of MASs to improve security and communicate with other agents through the MASs network in a complex environment. Primarily, adaptive neural networks are utilized to identify the unknown nonlinear dynamics parameter and ensure that the identification state errors converge asymptotically. Additionally, the optimized safe control design incorporates an actor–critic architecture based on reinforcement learning (RL) to approximate the Hamilton–Jacobi–Bellman (HJB) equation for nonlinear MASs. This distributed security control of MASs achieves optimal tracking using only local connectivity information. Finally, the proposed optimal algorithm for MASs safe formation-containment control is verified with MASs in an unknown obstacle environment.**

Impact Statement—**This work provides practical solutions for engineers controlling MASs like UAVs in complex environments. Adaptive neural networks adapt to unknown dynamics, while RL optimizes decision-making with limited local information, enhancing safety and efficiency. Future research should focus on enhancing robustness and applicability in real-world scenarios, developing strategies for effective MAS operation without external sensors, and improving decision-making in complex environments to solidify reliability and versatility in diverse applications.**

Index Terms—**Distributed safe control, formation-containment tracking, human-in-the-loop, nonlinear multi-agent systems (MASs), reinforcement learning (RL).**

I. Introduction

DISTRIBUTED multi-agent systems (MASs) and cooperative control have garnered increasing attention from researchers in recent decades due to their wide range of applications in various scenarios [1]. Recently, the formation containment problem, which originates from cooperative control, has been widely applied in reconnaissance [2] and systems scenarios [3]. As a crucial aspect of cooperative control, security consensus control has become a key consideration in practical applications, particularly [4]. Safe formation and containment controls build on the concepts of consensus control by addressing more specific coordination challenges [5].

Formation-containment control [6] is complex. Followers must gather inside a convex shape defined by the formation leaders [7], who must track a given path. Safety is crucial [8], especially when avoiding obstacles and no-fly zones in dangerous areas [9]. However, to the best of our knowledge, most existing formation containment control works do not consider the security of second-order nonlinear MASs with unknown dynamics. This makes these algorithms potentially inadequate for addressing an environment with unknown obstacles. The current formation-containment tracking control problems [10] have been primarily studied for linear systems and autonomous. However, there exist some occurrences due to the uncertainty and complexity of the work environment [11]. Therefore, developing control protocol to ensure that MASs can safely accomplish specific tasks in hazardous conditions is crucial. Meanwhile, existing obstacle avoidance algorithms rely on a priori knowledge [12], [13]. If not handled properly, this can lead to MASs colliding with obstacles, resulting in task failure. Human-interacted safe control has been introduced for formation-containment tracking in MASs. In this framework, a human operator can interact with MASs to improve the security of the network.

For unknown dynamics, recent studies have demonstrated the use of adaptive neural networks (NNs) for approximating unknown nonlinear dynamic equations [14]. One approach, the adaptive enclosing control method, has been proposed for nonlinear systems, incorporating state observer in the control process. However, this method heavily depends on precise system modeling, and any modeling errors may lead to instability in the system's control performance [15]. Another proposed

Received 20 October 2024; revised 19 February 2025; accepted 4 April 2025. Date of publication 11 April 2025; date of current version 31 August 2026. This work was supported by the Fundamental Research Funds for the Central Universities of China. This article was recommended for publication by Associate Editor Houbing Herbert Song upon evaluation of the reviewers' comments. *(Corresponding author: Jiang Zhao.)*

Luning Yang is with the School of Artificial Intelligence, Beihang University, Beijing 100191, China (e-mail: yangluning@buaa.edu.cn).

Pei Chi and Yingxun Wang are with the Institute of Unmanned Systems, Beihang University, Beijing 100191, China (e-mail: peichi@buaa.edu.cn; wangyx@buaa.edu.cn).

Jiang Zhao is with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China (e-mail: jzhao@buaa.edu.cn).

Digital Object Identifier 10.1109/TAI.2025.3559040

Symbol	Description
N	Set of nodes in the MASs.
E	Set of edges in the MASs.
A	Adjacency matrix of the MASs.
a_{ij}	Element of the adjacency matrix, indicating the connection between node i and node j.
L	Laplacian matrix, defined as $L = D - A$.
D	Degree matrix, with diagonal elements representing the degree of each node.
$x_0(t)$	Position of the tracking leader.
$v_0(t)$	Velocity of the tracking leader.
$f(v_0(t))$	Unknown nonlinear dynamics of the tracking leader.
$u_{\text{human}}(t)$	Control input from the human operator to the tracking leader.
$x_i(t)$	Position of the ith agent (formation leader or follower).
$v_i(t)$	Velocity of the ith agent (formation leader or follower).
$u_i(t)$	Control input of the ith agent.
$m_i(t)$	The state vector of the ith formation leader.
$\dot{m}_i(t)$	Derivative of the state vector of the ith formation leader.
$\delta x_i(t)$, $\delta v_i(t)$	Formation tracking position, velocity error of the ith agent.
$\xi x_i(t)$, $\xi v_i(t)$	Local containment position, velocity error of the ith follower.
$V_{E_i}(\delta_i(t), u_i(t))$	Cost function for the formation leaders and followers.
$H_E(\delta_j(t), u_i^*(t), V_i^*(t))$	HJB equation for the tracking leader and formation leaders.
$H_F(\delta_i(t), u_i^*(t), V_i^*(t))$	HJB equation for followers.
$\tilde{W}_i$	Weight error matrix of the neural network, defined as $\tilde{W}_i = W_i^* - \hat{W}_i$
$\hat{W}_i$	Estimated weight matrix of the neural network.
$S_i(v_i(t))$	Activation function of the neural network.
$b_i(v_i(t))$	Approximation error of the neural network.
τ_{x_i}, τ_{v_i}	Control parameters are used to design control input for formation leaders.
$\kappa_{x_i}, \kappa_{v_i}$	Control parameters are used to design control input for followers.
$\eta_{E_{ci}}, \eta_{F_{ci}}$	Learning rate parameters for updating the weight matrices for the critic network.
$\eta_{E_{ai}}, \eta_{F_{ai}}$	Learning rate parameters for updating the weight matrices for the actor network.
$u_{E_i}^*, u_{F_i}^*$	Optimal control inputs for formation and followers.
$\hat{W}_{E_{c_i}}, \hat{W}_{F_{c_i}}$	Weight matrices of the critic network for formation leaders and followers.
$\hat{W}_{E_{a_i}}, \hat{W}_{F_{a_i}}$	Weight matrices of the actor network for formation leaders and followers.
δ_i	The vector containing position and velocity control errors for formation leaders, defined as $\delta_i = [\delta_{x_i}, \delta_{v_i}]^T$.
ζ_i	The vector containing position and velocity control errors for followers is defined as $\zeta_i = [\zeta_{x_i}, \zeta_{v_i}]^T$.
c_i	The communication topology parameter is defined as $c_i = \sum_{j \in M \cup N} a_{ij} + a_{i0}$.
d_i	The communication topology parameter is defined as $d_i = \sum_{j \in M \cup N} a_{ij}$.
Δ_1, Δ_2	Constant terms in stability analysis.
κ, ι	Parameters in Lyapunov stability analysis.
$\lambda_{\min}$	Minimum eigenvalue of the matrix.

method, sliding mode control based on adaptive neural networks, aims to achieve zero tracking error convergence in a multiunmanned ship system comprising one leader and multiple followers. Despite its potential, this method faces a limitation in the slow learning speed of the global network [16]. Further research has explored adaptive local neural network fuzzy event triggering and clustering problems, suggesting a transmission control signal based on an event-triggering control strategy. However, balancing system performance with communication efficiency remains a challenge, and setting trigger conditions proves to be difficult. Adaptive neural networks exhibit strong nonlinear fitting capabilities, making them effective in handling unknown dynamics and uncertainties in nonlinear system control. However, their long training times, slow parameter convergence, and high computational complexity hinder their practical application in real-time control systems.

For environment with unknown obstacles, a model-independent Q-learning reinforcement learning method for MASs consistency control has been proposed. This approach does not rely on the environment's state or reward function, instead learning the optimal strategy through interaction with the environment. However, it requires large amounts of data to learn effective strategies, and the learning process is slow. Consequently, this method is not suitable for large-scale systems or complex environments [17]. Another method, based on policy gradients, addresses continuous action spaces and is well suited for MASs containment control, offering high flexibility. However, it suffers from high variance, low sample efficiency, and a tendency to converge to local optima in optimal control problems [18]. Existing reinforcement learning methods share common limitations, including the need for vast amounts of data and extensive interactions with the environment. This makes the learning process slow, particularly in safety-critical applications within complex environments. To overcome these challenges, we propose a MASs reinforcement learning method to achieve safe containment control with unknown obstacles.

Simultaneously, with the application of human interaction technologies, a human intervention method was proposed to prevent catastrophic behaviors by manually guiding the reinforcement learning process, ensuring that the MASs adopts safe and effective exploration strategies [19]. A distributed control method for human-in-the-loop autonomous systems was also investigated, focusing on the interaction between the uncertain system and the human operator [20]. Additionally, a distributed fault-tolerant tracking control problem for human-in-the-loop nonlinear leader-follower systems was addressed, with solutions for managing the interaction between the MASs and the human operator [21]. It is important to note that the autonomous systems studied in [22] did not account for the control inputs from the leader, limiting the applicability of these methods in controlling nonautonomous leaders. This limitation becomes particularly challenging in MASs involving second-order nonlinear dynamics. When human input is introduced, the system becomes a second-order nonlinear nonautonomous system, further complicating control efforts. To date, research in this area has been limited.

Additionally, safe formation containment control ensembles face significant challenges. Formation containment control faces significant challenges. One key challenge is solving the nonlinear Hamilton–Jacobi–Bellman (HJB) equations [23]. The complexity for optimal control arises because the MASs involved are nonlinear and not fully understood [24]. Additionally, obtaining global information in MASs presents another substantial challenge. The reinforcement learning control protocol was first proposed for single nonlinear systems [25]. Similarly, the optimal formation control protocols have been used to derive for unknown dynamics swarms in [26]. In addition, the linear quadratic regulator algorithm has been utilized for the linear MASs to obtain the optimal control protocol [27]. In the aforementioned work, the optimal containment control issue has been explored for high-order linear time-invariant MASs in [28]. Zhou and Tong [29] consider first-order nonlinear MASs formation containment control. Whereas, the optimal formation containment safe control problem for second-order systems has yet to be considered in the existing algorithms. Additionally, acquiring topological information in MASs is challenging, especially in large-scale networks [30], [31], [32]. The aforementioned works is inspired by the present study, which introduced an algorithm human-in-the-loop (HITLRL) that promotes nonlinear MASs with unknown dynamics safe tracking in the environment with unknown obstacles. The main contribution of the proposed HITLRL can be concluded as follows.

1) *Complex MASs Formation Containment Control*: First, this article settles the formation containment safe tracking control of second-order nonlinear unknown MASs more complex than the aforementioned work in [33] and [34] for the first time, while also minimizing the performance indicator.
2) *Human-in-the-Loop Safe Control*: Second, compared to the latest formation-containment study [35] and [37], the HITLRL algorithm is designed for nonautonomous MASs that can interact with a human operator. The human signal is accessible only to the tracking leader, and the algorithm enables safe tracking with human input.
3) *MASs Distributed Control*: Compared with the existing control protocols formation containment control protocols [38], [39], the proposed formation containment tracking protocol only relies on the limited information of MASs network topology, which is fully distributed. Furthermore, the separation principle is utilized to guarantee all signals of the MASs are asymptotically stable.

The rest of this article is organized as follows. Section II presents the preliminaries and problem formulation. In Section III, the main results are proposed. A simulation example is offered in Section IV. Finally, the conclusion is presented in Section V.

II. Preliminaries and Problem Formulation

A. Algebraic Graph Theory

In MASs, the communication relationships among agents can be represented using graphs. The communication topology can be described by a directed graph $G = \{V, E\}$, $V \in \{1, \ldots, N\}$ is the set of nodes, representing the agents. E is the set of edges, representing the communication links between agents. The Laplacian matrix is a key tool for describing graph structures, which is defined as $L = D - A$. D is the degree matrix, a diagonal matrix where D_{ii} represents the degree of node i, A is the adjacency matrix, where $A_{ij} = 1$ if $(i, j) \in E$, $A_{ij} = 0$ otherwise. The Laplacian matrix L describes the communication topology among agents. The off-diagonal elements L_{ij} represent the communication strength between agents i and j. The diagonal elements L_{ii} represent the degree of agent i. The second smallest eigenvalue λ_2 of the Laplacian matrix is known as the algebraic connectivity of the graph. It reflects the connectivity of the graph. A larger λ_2 indicates stronger connectivity, faster information propagation among agents, and faster system convergence. The largest eigenvalue λ_n of the Laplacian matrix reflects the expansion of the graph. A larger λ_n may lead to faster convergence but could also cause system instability.

B. Preliminaries and Problem Formulation

The MASs network includes a nonautonomous tracking leader, which can be interacted with by a human operator, N formation leaders, and M followers. The nonautonomous tracking leader providing a reference trajectory for the MASs is labeled by 0, the sequence $1, 2, \ldots, N$ denotes the tracking leader, the sequence $N+1, N+2, \ldots, N+M$ denotes the followers, which converge to a convex hull formed by the formation leaders.

We consider a continuous time-varying MASs and the tracking-leader nonlinear dynamics can be described by the model

$$\begin{cases} \dot{x}_0(t) = v_0(t) \\ \dot{v}_0(t) = f(v_0(t)) + u_{\text{human}}(t) \end{cases} \tag{1}$$

where x_0 indicates the position of the tracking leader (TL), v_0 is the velocity of the TL. f is the unknown nonlinear dynamics to be identified. $u_{\text{human}}(t)$ denotes the control input introduced by human operator. Additionally, the dynamics of formation-leaders and followers are as follows:

$$\begin{cases} \dot{x}_i(t) = v_i(t) \\ \dot{v}_i(t) = f(v_i(t)) + u_i(t), i \in \mathcal{M} \cup \mathcal{N} \end{cases} \tag{2}$$

where i denotes the ith index of formation leaders (FL) and followers, $x_i(t) \in \mathbb{R}^n$ is the position of leaders, and $u_i(t) \in \mathbb{R}^n$

signifies the control input for FL and followers.

Remark-1 (R.1): The HITLRL formation containment control strategy ensembles a human operator denoted as $u_{\mathrm{human}}(t)$ to adjust the trajectory of the MASs network directly. Additionally, the FL and followers agent can be influenced by the TL through the communication network.

Assumption-1 (A.1): It is assumed that there exists a minimum spanning tree in the graph. For each follower, there exists a directed path from at least a leader to it. In the case of a connected graph, the Laplacian matrix L has only one zero eigenvalue, and all other eigenvalues are positive. This ensures that information can be effectively transmitted among all agents.

Assumption-2 (A.2): There exists a Lipschitz that guarantees the consistent asymptotic convergence of the MASs. f_0, f_i ensures that the nonlinear dynamics have unique bounded initial MASs solution value. Specifically, there exists a Lipschitz $u_i \in U$ that guarantees the consistent asymptotic convergence of the MASs.

Assumption-3 (A.3): The states of the FL are represented by the vector $m_i(t) = col\,(m_1, \ldots, m_M)$ with $m_i(t)$, which is piecewise continuously differentiable. It will have broader practical application because of the time-varying formation and the nonzero derivative $\dot{m}_i(t)$ signal. To solve the difficulty, $\dot{m}(t)$ is assumed to be bounded.

Assumption-4 (A.4): The MASs achieve optimal safe time-varying formation containment tracking, ensuring that the states of the MASs, including leaders and followers, remain bounded.

Definition-1 (D.1): The second-order MASs achieve formation containment tracking supposing, if the following conditions are satisfied. The diagram of formation containment tracking is in Fig. 1.

1) The FL optimality of the MASs formation containment tracking control can be found such that

$$\begin{aligned} t \to \infty, &\|x_i(t) - m_{xi}(t) - x_0(t)\| \le e_{x1} \\ &\|v_i(t) - m_{vi}(t) - v_0(t)\| \le e_{v1}. \end{aligned} \tag{3}$$

2) The error e_2 of followers can be found that

$$t \to \infty, \left\| x_i(t) - \sum_{j \in \varepsilon} \gamma_{ij} x_j(t) \right\| \le e_2 \tag{4}$$

where $\|\xi_i(t)\| > 0$, $\|\delta_i(t)\| > 0$ satisfying $\sum_{j\in\varepsilon} \gamma_{ij} = 1$.

The formation tracking error corresponding to position and velocity as

$$\begin{aligned} \delta_{xi}(t) = &\sum_{i,j \in \mathcal{M}} a_{i,j} \left((x_i(t) - m_{xi}) - (x_j(t) - m_{xj}) \right) \\ &+ a_{i0} \left(x_i(t) - x_0(t) \right) \end{aligned} \tag{5}$$

$$\begin{aligned} \delta_{vi}(t) = &\sum_{i,j \in \mathcal{M}} a_{i,j} \left((v_i(t) - m_{vi}) - (v_j(t) - m_{vj}) \right) \\ &+ a_{i0} \left(v_i(t) - v_0(t) \right). \end{aligned} \tag{6}$$

Combining the dynamic and formation error, the local formation tracking position and velocity error is defined as

$$\begin{aligned} \dot{\delta}_{xi}(t) = &\sum_{i,j \in \mathcal{M}} a_{i,j} \left((\dot{x}_i(t) - \dot{m}_{xi}(t)) - (\dot{x}_j(t) - \dot{m}_{xj}(t)) \right) \\ &+ a_{i0} \left(\dot{x}_i(t) - \dot{m}_{xi}(t) - \dot{x}_0(t) \right) \\ = &\underbrace{\left(\sum_{i,j \in \mathcal{M}} a_{i,j} + a_{i0} \right)}_{c_i} \dot{x}_i(t) - c_i \dot{m}_{xi}(t) - \sum_{i,j \in \mathcal{M}} a_{i,j} \dot{x}_j(t) \\ &+ \sum_{i,j \in \mathcal{M}} a_{i,j} \dot{m}_{xj}(t) - a_{i0} \dot{x}_0(t) \end{aligned} \tag{7}$$

$$\begin{aligned} \dot{\delta}_{vi}(t) = &c_i f(v_i(t)) + c_i u_i(t) - c_i \dot{m}_{vi}(t) \\ &- \sum_{i,j \in \mathcal{N}} a_{i,j} f(v_j(t)) - \sum_{i,j \in \mathcal{N}} a_{i,j} u_j(t) \\ &+ \sum_{i,j \in \mathcal{N}} a_{i,j} \dot{m}_{vj}(t) - a_{i0} \left(f(v_0(t)) + u_{\mathrm{human}}(t) \right) \end{aligned} \tag{8}$$

For the follower i, the local containment error $\xi_{xi}(t)$, $\xi_{vi}(t)$ can be defined

$$\xi_{xi}(t) = \sum_{j \in \mathcal{M} \cup \mathcal{N}} a_{ij} \left(x_i(t) - x_j(t) \right) \tag{9}$$

$$\xi_{vi}(t) = \sum_{j \in \mathcal{M} \cup \mathcal{N}} a_{ij} \left(v_i(t) - v_j(t) \right). \tag{10}$$

The differentiation of the local containment error is defined as

$$\begin{aligned} \dot{\xi}_{xi}(t) &= \sum_{j \in \mathcal{M} \cup \mathcal{N}} a_{ij} \left(\dot{x}_i(t) - \dot{x}_j(t) \right) \\ &= \sum_{j \in \mathcal{M} \cup \mathcal{N}} a_{ij} \left(v_i(t) - v_j(t) \right) \end{aligned} \tag{11}$$

$$\begin{aligned} \dot{\xi}_{vi}(t) &= \sum_{j \in \mathcal{M} \cup \mathcal{N}} a_{ij} \left(\dot{v}_i(t) - \dot{v}_j(t) \right) \\ &= \sum_{j \in \mathcal{M} \cup \mathcal{N}} a_{ij} \left(f(v_i(t)) + u_i(t) - f(v_j(t)) - u_j(t) \right). \end{aligned} \tag{12}$$

The cost function of the MASs includes tracking leader and formation leader, which is defined as follows:

$$\begin{aligned} V_{Ei}\left(\delta_i(t), u_i(t) \right) = &\int_0^\infty {\delta_{xi}}^T(t) \delta_{xi}(t) dt \\ &+ \int_0^\infty {\delta_{vi}}^T(t) \delta_{vi}(t) dt + \int_0^\infty u_{E_i}^T(t) u_{E_i}(t) dt \end{aligned} \tag{13}$$

$$\begin{aligned} V_{Fj}\left(\xi_j(t), u_j(t) \right) = &\int_0^\infty {\xi_{xj}}^T(t) \xi_{xj}(t) dt \\ &+ \int_0^\infty {\xi_{vj}}^T(t) \xi_{vj}(t)\, dt + \int_0^\infty u_{F_j}^T(t) u_{F_j}(t)\, dt. \end{aligned} \tag{14}$$

Thus, the HJB equation of the tracking leader and formation leader

$$
\begin{aligned}
&H_{E_i}\left(\delta_i, u_i^*, V_i^*\right) = {\delta_{xi}}^T(t)\delta_{xi}\left(t\right) + {\delta_{vi}}^T(t)\delta_{vi}(t) \\
&+ u_i^T(t)u_i(t) + \frac{\partial V_E^T}{\partial \delta_{xi}} \begin{pmatrix} \underbrace{\left(\sum_{i,j\in\mathcal{M}} a_{i,j} + a_{i0}\right)}_{c_i} \dot{x}_i\left(t\right) \\ -c_i\dot{m}_i(t) - \sum_{i,j\in N} a_{i,j}\dot{x}_j(t) \\ + \sum_{i,j\in\mathcal{M}} a_{i,j}\dot{m}_j(t) \\ -a_{i0}\dot{x}_0(t) \end{pmatrix} \\
&+ \frac{\partial V_E^T}{\partial \delta_{vi}} \begin{pmatrix} c_i f\left(v_i(t)\right) + c_i u_i(t) - c_i\dot{m}_{vi}(t) \\ - \sum_{i,j\in\mathcal{N}} a_{i,j} f\left(v_j(t)\right) - \sum_{i,j\in\mathcal{N}} a_{i,j}u_j(t) \\ + \sum_{i,j\in\mathcal{N}} a_{i,j}\dot{m}_{vj}(t) - a_{i0}\begin{pmatrix} f\left(v_0(t)\right) \\ +u_{\text{human}}(t)\end{pmatrix} \end{pmatrix} = 0
\end{aligned} \tag{15}
$$

$$
\begin{aligned}
&H_{F_i}\left(\delta_i, u_i^*, V_i^*\right) = {\xi_{xi}}^T(t)\xi_{xi}(t) + {\xi_{vi}}^T(t)\xi_{vi}(t) \\
&+ u_i^T(t)u_i(t) + \frac{\partial V_F^T}{\partial \xi_{xi}} \left(\sum_{i,j\in\mathcal{N}} a_{ij}\left(v_i(t) - v_j(t)\right)\right) \\
&+ \frac{\partial V_F^T}{\partial \xi_{vi}} \left(\sum_{i,j\in N} a_{ij}\left(f\left(v_i(t)\right) + u_i\left(t\right) - f\left(v_j(t)\right) - u_j(t)\right)\right) \\
&= 0.
\end{aligned} \tag{16}
$$

Thus, the derivative for HJB equation of the tracking leader and formation leader

$$\frac{\partial H_{E_i}}{\partial u_i} = 2u_i(t) + \frac{\partial V_E^T}{\partial \delta_{vi}} c_i = 0 \tag{17}$$

$$\frac{\partial H_{F_i}}{\partial u_i} = 2u_i(t) + \frac{\partial V_F^T}{\partial \xi_{vi}} \left(\sum_{i,j\in\mathcal{M}\cup\mathcal{N}} a_{ij}\right) = 0. \tag{18}$$

The control laws for tracking leader and formation leader

$$u_i^* = -\frac{c_i}{2}\frac{dV_{Ei}^*}{d\delta_i}, i\in\mathcal{M} \tag{19}$$

$$u_i^* = -\frac{d_i}{2}\frac{dV_{Fi}^*}{d\xi_i}, i\in\mathcal{N} \tag{20}$$

where $m_{ij} = m_i - m_j$, $d_i = \sum_{i,j\in\mathcal{M}\cup\mathcal{N}} a_{i,j}$.

Problem of Interest: The main challenge in HITLRL for MASs is the complexity and nonlinearity of the systems, as well as limited information access. It is necessary to resolve several key issues, as follows.

1) How can we deploy safe formation tracking and containment simultaneously while minimizing the performance indicator for second-order MASs?
2) How can we guarantee the stability and convergence utilizing RL for human-interacting nonautonomous systems with time-varying formation leaders and followers?
3) How can we realize a fully distributed MASs network without relying on any global information about the network topology?

III. Main Results

In this section, we propose a safe HITLRL consisting of three specific components. First, NNs are utilized to identify the unknown second-order nonlinear dynamics of MASs in (2). Second, an actor–critic architecture is employed to achieve formation tracking error and containment error uniformly ultimately bounded (UUB). Finally, the HF operator is assisted to interact with MASs for safe formation-containment tracking. Fig. 2 illustrates the detailed architecture of safe HITLRL tracking controller via the NN-based actor–critic.

A. Identifier Design of Unknown Nonlinear Dynamics

The unknown nonlinear dynamics (2), which is accurately represented by the constructed NNs as

$$\dot{v}_i(t) = {W_i^*}^T S_i\left(v_i(t)\right) + u_i(t) + b_i\left(v_i\right), i\in\mathcal{M}\cup\mathcal{N} \tag{21}$$

where W_i^T denotes the estimation of the identifier estimated parameter. $\tilde{v}_i(t) = v_i(t) - \hat{v}_i(t)$ is the identifiable velocity error, and $\hat{v}_i(t)$ is the identification velocity.

Utilize the constructed NNs and rebuild the unknown nonlinear dynamics as

$$\dot{\tilde{v}}_i(t) = \tilde{W}_i^T S_i\left(v_i(t)\right) - \eta_s\tilde{v}_i(t) + b_i\left(v_i\right) \tag{22}$$

where $\tilde{W}_i^T = W_i^{*T} - \hat{W}_i^T$, $\eta_s > 0$ is a design parameter, and the weight estimates $\hat{W}_i$ are designed as

$$\dot{\hat{W}}_i(t) = S_i(v_i)\delta_{iv}^T - \hat{W}_i(t), i\in\mathcal{M}\cup\mathcal{N} \tag{23}$$

Remark-2 (R.2): In this article, the identifier error can converge to zero more than UUB. It allows for the subsequent design of an optimal formation-containment tracking control protocol based on a more accurate identification system.

B. HITLRL for MASs Formation-Containment Safe Control

The gradients of the value function can be divided into (24)–(25) to realize the formation containment control tracking

$$\frac{dV_{Ei}^*}{d\delta_i} = 2\frac{\tau_{xi}}{c_i}\delta_{xi}(t) + 2\frac{\tau_{vi}}{c_i}\delta_{vi}(t) + \frac{2}{c_i}f_i(x_i) + \frac{1}{c_i}V_{Ei}^0 \tag{24}$$

$$\frac{dV_{Fi}^*}{d\xi_i} = 2\frac{\kappa_{xi}}{d_i}\xi_{xi}(t) + 2\frac{\kappa_{vi}}{d_i}\xi_{vi}(t) + \frac{2}{d_i}f_i\left(x_i\right) + \frac{1}{d_i}V_{Fi}^0 \tag{25}$$

$$V_{Ei}^0 = W_{Ei}^{*T}S_{Ei}(\delta_i) + b_{Ei} \tag{26}$$

$$V_{Fi}^0 = W_{Fi}^{*T}S_{Fi}(\xi_i) + b_{Fi} \tag{27}$$

where W_{Ei}^{*T} and W_{Fi}^{*T} are the ideal weight matrices are Gaussian function. b_{Ei} and b_{Fi} are bounded approximation errors. Substituting (24)–(27) into (19)–(20), respectively, one can get the following:

$$
\begin{aligned}
u_{Ei}^* &= -\tau_{xi}\delta_{xi}(t) - \tau_{vi}\delta_{vi}(t) - f_i(x_i) \\
&\quad - \frac{1}{2}\left(W_{Ei}^{*T}S_{Ei}(\delta_i) + b_{Ei}\right), i\in\mathcal{M}
\end{aligned} \tag{28}
$$

$$
\begin{aligned}
u_{Fi}^* &= -\kappa_{xi}\xi_{xi}(t) - \kappa_{vi}\xi_{vi}(t) - f_i\left(x_i\right) \\
&\quad - \frac{1}{2}\left(W_{Fi}^{*T}S_{Fi}(\xi_i) + b_{Fi}\right), i\in\mathcal{N}
\end{aligned} \tag{29}
$$

where $\tau_{xi}, \tau_{vi}, \kappa_{xi}, \kappa_{vi} > 0$ is the continuous variables to be designed, $\delta_i = [\delta_{xi}, \delta_{vi}]^T$, $\xi_i = [\xi_{xi}, \xi_{vi}]^T$ is the column vector of tracking position and velocity.

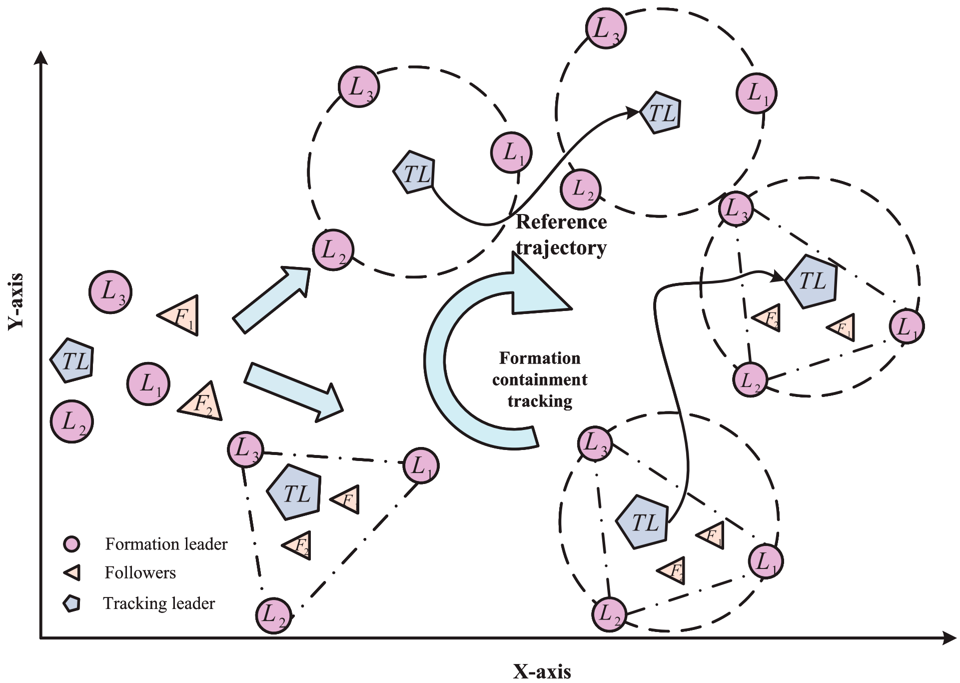

Fig. 1. Formation-containment tracking diagram.

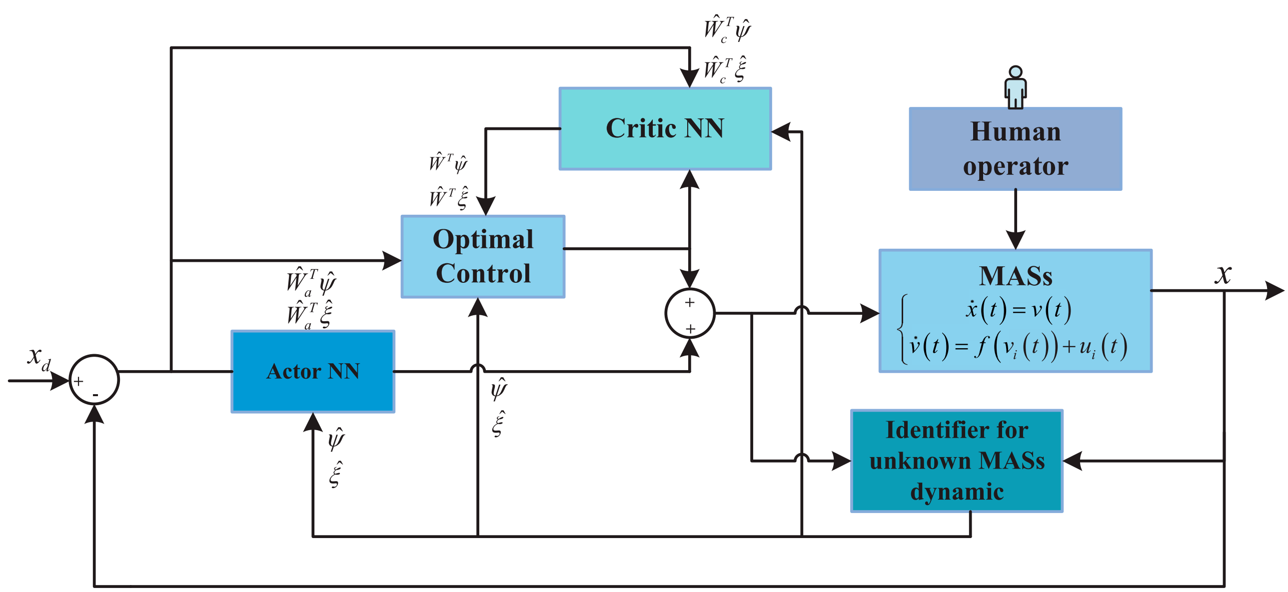

Fig. 2. Designed HITLRL algorithm for MASs structure.

However, the optimal control protocol is still including the unknown parameter W_{Ei}^*, W_{Fi}^* and the nonlinear dynamics f_i are unknown. Therefore, the actor–critic network is independent of the known system dynamics. Therefore, the critic network can be constructed as follows:

$$\frac{d\hat{V}_{Ei}^*}{d\delta_i} = 2\frac{\tau_{xi}}{c_i}\delta_{xi}(t) + 2\frac{\tau_{vi}}{c_i}\delta_{vi}(t) + \frac{2}{c_i}\hat{W}_i^T S_i\left(x_i(t)\right) + \frac{1}{c_i}\hat{W}_{Eci}^T S_{Ei}\left(\delta_i\right) \tag{30}$$

$$\frac{d\hat{V}_{Fi}^*}{d\xi_i} = 2\frac{\kappa_{xi}}{d_i}\xi_{xi}(t) + 2\frac{\kappa_{vi}}{d_i}\xi_{vi}(t) + \frac{2}{d_i}\hat{W}_i^T S_i\left(x_i(t)\right) + \frac{1}{d_i}\hat{W}_{Fci}^T S_{Fi}\left(\xi_i\right). \tag{31}$$

The critic NN weight matrix update laws are designed as

$$\dot{\hat{W}}_{Eci}(t) = -S_{Ei}\left(\delta_i\right)\delta_{xi}(t) - S_{Ei}\left(\delta_i\right)\delta_{vi}(t) - \eta_{Eci}S_{Ei}\left(\delta_i\right)S_{Ei}^T\left(\delta_i\right)\hat{W}_{Eci} \tag{32}$$

$$\dot{\hat{W}}_{Fci}(t) = -S_{Fi}(\xi_i)\,\xi_{xi}(t) - S_{Fi}(\xi_i)\,\xi_{vi}(t) - \eta_{Fci}S_{Fi}(\xi_i)\,S_{Fi}^T(\xi_i)\,\hat{W}_{Fci} \quad (33)$$

where η_{Eci}, η_{Fci} are designed positive parameters. The actor network can be constructed as follows:

$$u_{Ei} = -\tau_{xi}\delta_{xi}(t) - \tau_{vi}\delta_{vi}(t) - \hat{W}_i^T S_i(x_i) - \frac{1}{2}\hat{W}_{Eai}^T S_{Ei}(\delta_i), i \in \mathcal{M} \quad (34)$$

$$u_{Fi} = -\kappa_{xi}\xi_{xi}(t) - \kappa_{vi}\xi_{vi}(t) - \hat{W}_i^T S_i(v_i) - \frac{1}{2}\hat{W}_{Fai}^T G_{Fi}(\xi_i), i \in \mathcal{N} \quad (35)$$

where $\hat{W}_{Eai}^T$, $\hat{W}_{Eci}^T$ are the actor NN weight matrices. The actor NN weight matrices update laws are designed as

$$\dot{\hat{W}}_{Eai}(t) = -S_{Ei}(\delta_i)\,S_{Ei}^T(\delta_i)\begin{pmatrix}\eta_{Eai}\left(\hat{W}_{Eai} - \hat{W}_{Eci}\right)\\ +\eta_{Eci}\hat{W}_{Eci}\end{pmatrix} \quad (36)$$

$$\dot{\hat{W}}_{Fai}(t) = -S_{Fi}(\xi_i)\,S_{Fi}^T(\xi_i)\begin{pmatrix}\eta_{Fai}\left(\hat{W}_{Fai} - \hat{W}_{Fci}\right)\\ +\eta_{Fci}\hat{W}_{Fci}\end{pmatrix} \quad (37)$$

Theorem-1 (Th. 1): Consider a MASs generated in (1) and assume that Assumptions A.1–A.4 hold, the safe reinforcement learning algorithm interacted with human operator in (18) is implemented by the NN-based RL. The actor–critic NN are updated according to (21)–(22), and the design parameters satisfy: $\tau_{xi} \geq (9/4)$, $\tau_{vi} \geq (15/4)$, $\eta_{Eci} > \max\{1, 2\eta_{Eai}\}$, $\kappa_{vi} > (\tau_{vi}/2) + (5/2)$, $\kappa_{xi} > (\tau_{xi}/2) + (5/2)$, $\eta_{Eai} > 0$.

Then, the following objectives can be achieved.

1) These errors $\delta_{xi}(t)$, $\delta_{vi}(t)$, $\xi_{xi}(t)$, $\xi_{vi}(t)$, $\hat{W}_{Eai}(t)$, $\hat{W}_{Eci}(t)$, $\hat{W}_{Fai}(t)$, $\hat{W}_{Fci}(t)$ are semiglobally UUB.
2) MASs output follows the predefined tracking leader of $x_0(t)$ on the desired trajectory.

Proof: The stability of the safe formation-containment tracking control can be proved from formation leaders and followers, respectively.

1) Step-1: Stability for Formation-Leaders: Choosing the following Lyapunov function candidate for formation-leaders:

$$L_1 = \frac{1}{2}\sum_{i\in\mathcal{M}}\delta_i^T\delta_i + \frac{1}{2}\sum_{i\in\mathcal{M}}Tr\left(\tilde{W}_i^T\tilde{W}_i\right) + \frac{1}{2}\sum_{i\in\mathcal{M}}Tr\left(\tilde{W}_{Eci}^T\tilde{W}_{Eci}\right) + \frac{1}{2}\sum_{i\in\mathcal{M}}Tr\left(\tilde{W}_{Eai}^T\tilde{W}_{Eai}\right) \quad (38)$$

The differential of the Lyapunov function is as follows:

$$\dot{L}_1 = \sum_{i\in\mathcal{M}}\delta_i^T\dot{\delta}_i + \sum_{i\in\mathcal{M}}Tr\left(\tilde{W}_i^T\dot{\hat{W}}_i\right) + \sum_{i\in\mathcal{M}}Tr\left(\tilde{W}_{Eci}^T\dot{\hat{W}}_{Eci}\right) + \sum_{i\in\mathcal{M}}Tr\left(\tilde{W}_{Eai}^T\dot{\hat{W}}_{Eai}\right). \quad (39)$$

Substitute $\dot{\delta}_{xi}(t)$,$\dot{\delta}_{vi}(t)$,$\dot{\hat{W}}_{Eai}(t)$, $\dot{\hat{W}}_{Eai}(t)$ into the Lyapunov function and utilize the constructed NN parameters to represent the error expression

$$\begin{aligned}\dot{L}_1(t) &= \sum_{i=1}^{M}\delta_{xi}^T(t)\begin{pmatrix}v - \tau\delta_{xi}(t) - \frac{1}{2}\hat{W}_{Eai}^T S_{Ei}(\delta_{xi})\\ -\dot{m}_{xi}(t) - f_0(x_0)\end{pmatrix}\\ &+ \sum_{i=1}^{M}\delta_{vi}^T(t)\begin{pmatrix}\tilde{W}_i^T S_i(v) - \tau\delta_{vi}(t) - \frac{1}{2}\hat{W}_{Eai}^T S_{Ei}(\delta_{vi})\\ +b_i - \dot{m}_{vi}(t) - f_0(x_0)\end{pmatrix}\\ &+ \sum_{i=1}^{M}Tr\left\{\tilde{W}_i^T\left(S_i(v_i)\delta_i - \hat{W}_i\right)\right\}\\ &- \sum_{i=1}^{M}\eta_{Eci}Tr\left\{\tilde{W}_{Eci}^T S_{Ei}(\delta_i)\,S_{Ei}^T(\delta_i)\,\hat{W}_{Eci}\right\}\\ &+ \sum_{i=1}^{M}Tr\left\{\tilde{W}_{Eai}^T S_{Ei}(\delta_i)\,S_{Ei}^T(\delta_i)\begin{pmatrix}\eta_{Eai}\left(\hat{W}_{Eai} - \hat{W}_{Eci}\right)\\ +\eta_{Eci}\hat{W}_{Eci}\end{pmatrix}\right\}.\end{aligned} \quad (40)$$

The first term can be simplified into

$$\begin{aligned}l_{11} &= \delta_{xi}^T(t)\begin{pmatrix}v - \tau\delta_{xi}(t) - \frac{1}{2}\hat{W}_{Eai}^T S_{Ei}(\delta_{xi})\\ -\dot{m}_{xi}(t) - f_0(x_0)\end{pmatrix}\\ &\leq \frac{7-4\tau}{4}\delta_{xi}^T\delta_{xi} + \frac{1}{2}v_i^T v_i\\ &+ \frac{1}{4}Tr\left(\hat{W}_{Eai}^T S_{Ei}(\delta_{vi})S_{Ei}^T(\delta_{vi})\hat{W}_{Eai}\right)\\ &+ \frac{1}{2}\dot{m}_{xi}^T\dot{m}_{xi} + \frac{1}{2}f_0^T f_0.\end{aligned} \quad (41)$$

The second term can be simplified into

$$\begin{aligned}l_{12} &= \delta_{vi}^T\begin{pmatrix}\tilde{W}_i^T S_i - \tau\delta_{vi} - \frac{1}{2}\hat{W}_{Eai}^T S_{Ei} + b_i\\ -\dot{m}_i(t) - f_0(x_0) - u_{\text{human}}\end{pmatrix}\\ &\leq \frac{11-4\tau}{4}\delta_{vi}^T\delta_{vi} + \frac{1}{2}Tr\left(\tilde{W}_i^T S_i(v)\,S_i{}^T(v)\,\tilde{W}_i\right)\\ &+ \frac{1}{4}Tr\left(\hat{W}_{Eai}^T S_{Ei}(\delta_{vi})S_{Ei}^T(\delta_{vi})\hat{W}_{Eai}\right)\\ &+ \frac{1}{2}\dot{m}_i^T\dot{m}_i + \frac{1}{2}f_0{}^T f_0 + \frac{1}{2}b^T b + \frac{1}{2}u_{\text{human}}^T u_{\text{human}}.\end{aligned} \quad (42)$$

The third term can be simplified by Young's inequalities

$$\begin{aligned}l_{13} &= Tr\left\{\tilde{W}_i^T\left(S_i(v_i)\delta_{vi} - \hat{W}_i(t)\right)\right\}\\ &= Tr\left(\tilde{W}_i^T S_i(v_i)\delta_{vi}\right) - Tr\left(\tilde{W}_i^T\hat{W}_i\right)\\ &\leq \frac{1}{2}Tr\left(\tilde{W}_i^T S_i(v_i)S_i^T(v_i)\tilde{W}_i\right)\\ &+ \frac{1}{2}\delta_{vi}{}^T\delta_{vi} - Tr\left(\left(\hat{W}_i^T - W_i^*\right)\hat{W}_i\right)\\ &\leq \frac{1}{2}Tr\left(\tilde{W}_i^T S_i(v_i)S_i^T(v_i)\tilde{W}_i\right) + \frac{1}{2}\delta_{vi}{}^T\delta_{vi}\\ &- \begin{pmatrix}\frac{1}{2}Tr\left(\tilde{W}_i^T(v_i)\tilde{W}_i(v_i)\right) + \frac{1}{2}Tr\left(\hat{W}_i^T(v_i)\hat{W}_i(v_i)\right)\\ -\frac{1}{2}\left(W_i{}^*(v_i)W_i{}^*(v_i)\right)\end{pmatrix}\end{aligned} \quad (43)$$

The fourth term can be simplified by Young's inequalities

$$\begin{aligned}l_{14} &= -\tilde{W}_{Eci}^T S_{Ei}(\delta_i)\,\delta_{xi}(t) - \tilde{W}_{Eci}^T S_{Ei}(\delta_i)\,\delta_{vi}(t)\\ &- \eta_{Eci}Tr\left\{\tilde{W}_{Eci}^T S_{Ei}(\delta_i)S_{Ei}^T(\delta_i)\hat{W}_{Eci}\right\}\\ &\leq \frac{1}{2}Tr\left\{\tilde{W}_{Eci}^T S_{Ei}(\delta_i)S_{Ei}^T(\delta_i)\tilde{W}_{Eci}\right\} + \frac{1}{2}\delta_{xi}^T\delta_{xi}\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{2}Tr\left\{\tilde{W}_{Eci}^{T}S_{Ei}(\delta_i)S_{Ei}^{T}(\delta_i)\tilde{W}_{Eci}\right\}+\frac{1}{2}\delta_{vi}^{T}\delta_{vi}\\
&-\frac{1}{2}\eta_{Eci}Tr\left\{\tilde{W}_{Eci}^{T}S_{Ei}(\delta_i)S_{Ei}^{T}(\delta_i)\tilde{W}_{Eci}\right\}\\
&-\frac{1}{2}\eta_{Eci}Tr\left\{\hat{W}_{Eci}^{T}S_{Ei}(\delta_i)S_{Ei}^{T}(\delta_i)\hat{W}_{Eci}\right\}\\
&+\frac{1}{2}\eta_{Eci}Tr\left\{W_{Eci}^{*T}S_{Ei}(\delta_i)S_{Ei}^{T}(\delta_i)W_{Eci}^{*}\right\}.
\end{aligned}
\tag{44}
$$

The fifth term can be simplified by Young's inequalities

$$
\begin{aligned}
l_{15}&=Tr\left(\tilde{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\begin{pmatrix}\eta_{Eai}\left(\hat{W}_{Eai}-\hat{W}_{Eci}\right)\\+\eta_{Eci}\hat{W}_{Eci}\end{pmatrix}\right)\\
&=Tr\begin{pmatrix}\eta_{Eai}\tilde{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eai}\\-\eta_{Eai}\tilde{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eci}\\+\eta_{Eci}\tilde{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eci}\end{pmatrix}\\
&\le\frac{1}{2}\eta_{Eai}Tr\left(\tilde{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\tilde{W}_{Eai}\right)\\
&\quad+\frac{1}{2}\eta_{Eai}Tr\left(\hat{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eai}\right)\\
&\quad-\frac{1}{2}\eta_{Eai}Tr\left(\hat{W}_{Eai}^{*}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eai}^{*}\right)\\
&\quad+\frac{\eta_{Eci}+\eta_{Eai}}{2}Tr\left(\tilde{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\tilde{W}_{Eai}\right)\\
&\quad+\frac{\eta_{Eci}+\eta_{Eai}}{2}Tr\left(\hat{W}_{Eci}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eci}\right)
\end{aligned}
\tag{45}
$$

Thus,

$$
\begin{aligned}
\dot{L}(t)\le&-\sum_{i\in\mathcal{M}}\frac{4\tau_{xi}-9}{4}\delta_{xi}^{T}\delta_{xi}\\
&-\sum_{i\in\mathcal{M}}\frac{4\tau_{vi}-15}{4}\delta_{vi}^{T}\delta_{vi}\\
&-\sum_{i\in\mathcal{M}}\left(-\frac{3}{2}\right)Tr\left(\tilde{W}_i^{T}\tilde{W}_i\right)\\
&-\sum_{i\in\mathcal{M}}\left(-1+\frac{1}{2}\eta_{Eci}\right)\tilde{W}_{Eci}^{T}S_{Ei}\left(\delta_i\right){S_{Ei}}^{T}\left(\delta_i\right)\tilde{W}_{Eci}\\
&-\sum_{i\in\mathcal{M}}\left(\frac{-2\eta_{Eai}-\eta_{Eci}}{2}\right)Tr\left(\tilde{W}_{Eai}^{T}S_{Ei}(\delta_i){S_{Ei}}^{T}(\delta_i)\tilde{W}_{Eai}\right)\\
&+\Delta_1
\end{aligned}
\tag{46}
$$

which can be bounded by a constant Δ, $\lambda_{\min}^{s_i}$ is the minimal eigenvalue of $S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)$

$$
\begin{aligned}
\Delta_1=&\frac{1}{2}v^{T}v+2Tr\left(\hat{W}_{Eai}^{T}S_{Ei}S_{Ei}^{T}\hat{W}_{Eai}\right)+\frac{1}{2}\dot{m}_{xi}^{T}\dot{m}_{xi}+f_0^{T}f_0\\
&+u_{\text{human}}^{T}u_{\text{human}}+\frac{1}{2}\dot{m}_{vi}^{T}\dot{m}_{vi}+\frac{1}{2}b^{T}b\\
&+\frac{1}{2}Tr\hat{W}_i^{T}(v_i)\hat{W}_i(v_i)\\
&+\frac{\eta_{Eai}+\eta_{Eci}-1}{2}Tr\left(\hat{W}_{Eci}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eci}\right)\\
&+\frac{\eta_{Eai}+1}{2}Tr\left(\hat{W}_{Eai}^{T}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eai}\right)\\
&-Tr\left(\frac{1}{2}\eta_{Eai}\hat{W}_{Eai}^{*}S_{Ei}\left(\delta_i\right)S_{Ei}^{T}\left(\delta_i\right)\hat{W}_{Eai}^{*}\right)\\
&+\frac{1}{2}\eta_{Eci}Tr\left(W_{Eci}^{*T}S_{Ei}(\delta_i)S_{Ei}^{T}(\delta_i)W_{Eci}^{*}\right)\\
&-\frac{1}{2}\left(\hat{W}_i^{*}(v_i)\hat{W}_i^{*}(v_i)\right).
\end{aligned}
\tag{47}
$$

Let $\beta=\min\left\{(4\tau_{xi}-9/4),(4\tau_{vi}-15/4),-(3/2),-1+(1/2)\eta_{Eci},(-2\eta_{Eai}-\eta_{Eci}/2)\right\}$, the (46) becomes the following one:

$$
\dot{L}\le-\beta L+\Delta.
\tag{48}
$$

The following inequality can be held:

$$
L(t)\le e^{-\beta t}L\left(0\right)+\frac{\Delta_1}{\beta}\left(1-e^{-\beta t}\right).
\tag{49}
$$

2) Step-2: Stability for followers: Consider the following Lyapunov function about followers:

$$
\begin{aligned}
L_2=&\frac{1}{2}\sum_{i\in\mathcal{N}}\xi_i^{T}\xi_i+\frac{1}{2}\sum_{i\in\mathcal{N}}Tr\left(\tilde{W}_i^{T}\tilde{W}_i\right)\\
&+\frac{1}{2}\sum_{i\in\mathcal{N}}Tr\left(\tilde{W}_{Fai}^{T}\tilde{W}_{Fai}\right)+\frac{1}{2}\sum_{i\in\mathcal{N}}Tr\left(\tilde{W}_{Fci}^{T}\tilde{W}_{Fci}\right).
\end{aligned}
\tag{50}
$$

Thus, the derivative of (5) is described by

$$
\begin{aligned}
\dot{L}_2=&\sum_{i\in\mathcal{N}}\xi_{xi}^{T}(t)\left(\sum_{j\in\mathcal{M}\cup\mathcal{N}}a_{ij}\left(v_i(t)-v_j(t)\right)\right)\\
&+\sum_{i\in\mathcal{N}}\xi_{vi}^{T}(t)\left(\sum_{j\in\mathcal{N}}a_{ij}\begin{pmatrix}f\left(v_i(t)\right)+u_i\left(t\right)\\-f\left(v_j(t)\right)+u_j(t)\end{pmatrix}\right)\\
&+\sum_{i\in\mathcal{N}}Tr\left(\tilde{W}_{Fai}^{T}\begin{pmatrix}-S_{Fi}\left(\xi_i\right)S_{Fi}^{T}\left(\xi_i\right)\\\times\begin{pmatrix}\eta_{Fai}\left(\hat{W}_{Fai}-\hat{W}_{Fci}\right)\\+\eta_{Fci}\hat{W}_{Fci}\end{pmatrix}\end{pmatrix}\right)\\
&+\sum_{i\in\mathcal{N}}Tr\left(\tilde{W}_{Fci}^{T}\begin{pmatrix}-\tau_{xi}\delta_{xi}(t)-\tau_{vi}\delta_{vi}(t)\\-\eta_{Fci}S_{Fi}\left(\xi_i\right)S_{Fi}^{T}\left(\xi_i\right)\hat{W}_{Fci}\end{pmatrix}\right).
\end{aligned}
\tag{51}
$$

The first term can be simplified

$$
\begin{aligned}
l_{21}&=\sum_{i\in\mathcal{N}}\xi_{xi}^{T}\left(d_iv_i(t)-v_j(t)\right)\\
&=\sum_{i\in\mathcal{N}}\xi_{xi}^{T}\sum_{i\in\mathcal{M}\cup\mathcal{N}}\left(v_i(t)-v_j(t)\right)\\
&\le\sum_{i\in\mathcal{N}}\frac{d_i}{2}\xi_{xi}^{T}\xi_{xi}+\frac{d_i}{2}v_i^{T}v_i+\frac{d_i}{2}\xi_{xi}^{T}\xi_{xi}+\frac{d_i}{2}v_j^{T}v_j.
\end{aligned}
\tag{52}
$$

The second term can be simplified

$$
\begin{aligned}
l_{22}=&\sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}}a_{ij}\begin{pmatrix}\xi_{vi}^{T}(t)\tilde{W}_{fi}^{T}S_{fi}(v_i)\\-a_i\xi_{vi}^{T}(t)\xi_{vi}(t)\\-\frac{1}{2}\xi_{vi}^{T}(t)\hat{W}_{Fai}^{T}G_{Fi}(\xi_{vi})\\+\xi_{vi}^{T}(t)b_i(v_i)\end{pmatrix}\\
&-\sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}}a_{ij}\begin{pmatrix}\xi_{vi}^{T}\tilde{W}_{fj}^{T}S_{fj}(v_j)-\kappa_{vj}\xi_{vi}^{T}(t)\xi_{vj}(t)\\-\kappa_{xj}\xi_{vi}^{T}(t)\xi_{xj}(t)\\-\frac{1}{2}\xi_{vi}^{T}(t)\hat{W}_{Eaj}^{T}G_{Ei}(\xi_{vj})\\+\xi_{vi}^{T}(t)b_j(v_j)\end{pmatrix}
\end{aligned}
\tag{53}
$$

$$l_{22} \leq \sum_{i\in\mathcal{N}} \begin{pmatrix} \sum_{j\in\mathcal{M}} \frac{1}{4} Tr\left(\hat{W}_{Fai}^T S_{Fi}(\xi_i) S_{Fi}^T(\xi_i(t)) \hat{W}_{Fai}(t)\right) \\ + \sum_{j\in\mathcal{M}} a_{ij} \left(\frac{1}{2} Tr\begin{pmatrix} \tilde{W}_{fi}^T(t) S_{fi}(v_i(t)) \times \\ S_{fi}^T(v_i(t)) \tilde{W}_{fi}(t) \end{pmatrix}\right) \\ + \sum_{j\in\mathcal{M}} a_{ij}\left(-\kappa_{vi} + \frac{\tau_{vi}}{2} + \frac{5}{2}\right)\xi_{vj}^T(t)\xi_{vj}(t) \\ + \sum_{j\in\mathcal{M}} a_{ij}\left(-\kappa_{xi} + \frac{\tau_{xi}}{2} + \frac{5}{2}\right)\xi_{xj}^T(t)\xi_{xj}(t) \end{pmatrix} + K \tag{54}$$

$$\begin{aligned} K = & \sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}} a_{ij}\left(\frac{(\delta_i)}{2}\tilde{v}_i^T(t)\tilde{v}_i(t)\right) \\ & + \sum_{i\in\mathcal{F}}\sum_{j\in\mathcal{E}} a_{ij}\frac{\delta_j}{2}\tilde{v}_j^T(t)\tilde{v}_j(t) \\ & + \sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}} a_{ij}\frac{\tau_{xj}}{2}\xi_{xj}^T(t)\xi_{xj}(t) \\ & + \sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}} a_{ij}\frac{\tau_{vj}}{2}\xi_{vj}^T(t)\xi_{vj}(t) \\ & + \sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}} a_{ij}\frac{1}{2}Tr\begin{pmatrix}\tilde{W}_{fj}^T(t)S_{fj}(v_i(t)) \\ \times S_{fj}^T(v_i(t))\tilde{W}_{fi}(t)\end{pmatrix} \\ & + \sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{M}} a_{ij}\frac{1}{4}Tr\begin{pmatrix}\hat{W}_{Eaj}^T S_{Ej}(\xi_j) \\ \times S_{Ej}^T(\xi_j)\hat{W}_{Eaj}(t)\end{pmatrix}. \end{aligned} \tag{55}$$

The third term can be simplified

$$\begin{aligned} l_{23} &= Tr\left\{\tilde{W}_i^T\left(S_i(v_i)\xi_{vi} - \hat{W}_i(t)\right)\right\} \\ &= Tr\left(\tilde{W}_i^T S_i(v_i)\xi_{vi}\right) - Tr\left(\tilde{W}_i^T\hat{W}_i\right) \\ &\leq \frac{1}{2}Tr\left(\tilde{W}_i^T S_i(v_i)S_i^T(v_i)\tilde{W}_i\right) + \frac{1}{2}{\xi_{vi}}^T\xi_{vi} \\ &\quad - Tr\left(\left(\hat{W}_i^T - W_i^*\right)\hat{W}_i\right) \\ &\leq \frac{1}{2}Tr\left(\tilde{W}_i^T S_i(v_i)S_i^T(v_i)\tilde{W}_i\right) + \frac{1}{2}{\xi_{vi}}^T\xi_{vi} \\ &\quad - \begin{pmatrix}\frac{1}{2}Tr\tilde{W}_i^T(v_i)\tilde{W}_i(v_i) + \frac{1}{2}Tr\hat{W}_i^T(v_i)\hat{W}_i(v_i) \\ -\frac{1}{2}{W_i}^*(v_i){W_i}^*(v_i)\end{pmatrix}. \end{aligned} \tag{56}$$

The fourth term can be simplified

$$\begin{aligned} l_{24} \leq & \frac{\eta_{Fai}}{2}Tr\left(\tilde{W}_{Fai}^T S_{Fi}(\xi_i) S_{Fi}(\xi_i)\tilde{W}_{Fai}\right) \\ & + \frac{\eta_{Fai}}{2}Tr\left(\hat{W}_{Fai}^T S_{Fi}(\xi_i) S_{Fi}^T(\xi_i)\hat{W}_{Fai}\right) \\ & + \frac{\eta_{Fai}}{2}Tr\left(\tilde{W}_{Fai}^T S_{Fi}(\xi_i) S_{Fi}^T(\xi_i)\tilde{W}_{Fai}\right) \\ & + \frac{\eta_{Fai}}{2}Tr\left(\hat{W}_{Fci}^T S_{Fi}(\xi_i) S_{Fi}^T(\xi_i)\hat{W}_{Fci}\right) \\ & + \frac{\eta_{Fci}}{2}Tr\left(\tilde{W}_{Fai}^T S_{Fi}(\xi_i) S_{Fi}^T(\xi_i)\tilde{W}_{Fai}\right) \\ & + \frac{\eta_{Fci}}{2}Tr\left(\tilde{W}_{Fci}^T S_{Fi}(\xi_i) S_{Fi}^T(\xi_i)\tilde{W}_{Fci}\right) \end{aligned} \tag{57}$$

The fifth term can be simplified

$$\begin{aligned} l_{25} &= Tr\left(\tilde{W}_{Fci}^T\begin{pmatrix}-S_{Fi}(\xi_i)\xi_{xi}(t) - S_{Fi}(\xi_i)\xi_{vi}(t) \\ -\eta_{Fci}S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\hat{W}_{Fci}\end{pmatrix}\right) \\ &\leq \frac{1}{2}Tr\left(\tilde{W}_{Fci}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\tilde{W}_{Fci}\right) + \frac{1}{2}\xi_{xi}^T\xi_{xi} \\ &\quad + \frac{1}{2}Tr\left(\tilde{W}_{Fci}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\tilde{W}_{Fci}\right) + \frac{1}{2}\xi_{vi}^T\xi_{vi} \\ &\quad + \frac{\eta_{Fci}}{2}Tr\left(\tilde{W}_{Fci}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\tilde{W}_{Fci}\right) \\ &\quad + \frac{\eta_{Fci}}{2}Tr\left(\hat{W}_{Fci}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\hat{W}_{Fci}\right). \end{aligned} \tag{58}$$

Synthesize (52)–(55) to simplify

$$\begin{aligned} \dot{L} \leq & \sum_{i\in\mathcal{N}}\begin{pmatrix}\sum_{j\in\mathcal{M}} a_{ij}\left(-\kappa_{vi} + \frac{\tau_{vi}}{2} + \frac{5}{2}\right)\xi_{vj}^T(t)\xi_{vj}(t) \\ +{\xi_{vi}}^T\xi_{vi} \\ + \sum_{j\in\mathcal{M}} a_{ij}\begin{pmatrix}\left(-\kappa_{xi} + \frac{\tau_{xi}}{2} + \frac{5}{2}\right)\xi_{xj}^T(t)\xi_{xj}(t) \\ +\frac{1}{2}\xi_{xi}^T\xi_{xi}\end{pmatrix}\end{pmatrix} \\ & + \sum_{j\in\mathcal{M}} a_{ij}\begin{pmatrix}Tr\begin{pmatrix}\tilde{W}_i^T(t)S_i(v_i(t)) \times \\ S_i^T(v_i(t))\tilde{W}_i(t)\end{pmatrix} \\ -Tr\tilde{W}_i^T(v_i)\tilde{W}_i(v_i)\end{pmatrix} \\ & + \frac{2\eta_{Fai} + \eta_{Fci}}{2}Tr\left(\tilde{W}_{Fai}^T S_{Fi}(\xi_i)S_{Fi}(\xi_i)\tilde{W}_{Fai}\right) \\ & + \frac{2 + 2\eta_{Fci}}{2}Tr\left(\tilde{W}_{Fci}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\tilde{W}_{Fci}\right) + K + \Delta_2 \end{aligned} \tag{59}$$

where

$$\begin{aligned} \Delta_2 = & \sum_{i\in\mathcal{N}}\begin{pmatrix}\sum_{j\in\mathcal{M}}\left(\frac{1}{4} + \frac{\eta_{Fai}}{2}\right) \\ \times Tr\left(\hat{W}_{Fai}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i(t))\hat{W}_{Fai}(t)\right)\end{pmatrix} \\ & + \frac{\eta_{Fai} + \eta_{Fci}}{2}Tr\left(\hat{W}_{Fci}^T S_{Fi}(\xi_i)S_{Fi}^T(\xi_i)\hat{W}_{Fci}\right) \\ & + \sum_{i\in\mathcal{N}}\xi_{xi}^T\sum_{i\in\mathcal{M}\cup\mathcal{N}}(v_i(t) - v_j(t)) - \frac{1}{2}{W_i}^*(v_i){W_i}^*(v_i). \end{aligned} \tag{60}$$

Let $\kappa = \min\left\{\begin{matrix}-\kappa_{vi}+(\tau_{vi}/2)+(5/2), -\kappa_{xi}+(\tau_{xi}/2)+3, \\ -1 + \lambda_{\min}\left(S_i(x_i)\,{S_i}^T(x_i)\right), \\ (2\eta_{Fai} + \eta_{Fci}/2)(2 + 2\eta_{Fci}/2)\end{matrix}\right\}$.
Thus, the Lyapunov can be proved

$$\dot{L} \leq -\kappa L + \iota \tag{61}$$

$$L \leq e^{-\kappa t}V(0) + \frac{\iota}{\kappa}\left(1 - e^{-\kappa t}\right). \tag{62}$$

The above inequality shows that the tracking error can converge to the desired precision. Thus, Th.1 is proved.

IV. Simulation

The proposed HITLRL protocol algorithm is verified by a group of fifteen unmanned aerial vehicles (UAVs), which includes one tracking leader, six formation leaders and eight followers. The connection topology is shown in Fig. 3. The MASs dynamic of FL and followers can be described in the following:

$$\begin{cases}\dot{x}_i(t) = v_i(t) \\ \dot{v}_i(t) = \begin{bmatrix}\alpha_i x_{i1}\cos(x_{i1}\cdot v_{i1}) \\ \beta_i x_{i2}\sin(x_{i2}\cdot v_{i2})\end{bmatrix} + u_i,\ i = 1,\ldots,14\end{cases} \tag{63}$$

where $\alpha_{\mathrm{i}=1,\ldots,14} = 0.7, 0.3, 0.8, 0.5, 0.7, 0.7, 0.5, 0.7, 0.8, 0.72, 0.79, 0.85, 0.74, 0.76$ and $\beta_{\mathrm{i}=1,\ldots,14} = 1.5, 1.4, 2, 1.5, 1.5, 1.5, 1.7, 1.8, 1.6, 1.62, 1.61, 1.52, 1.56, 1.52$, respectively.

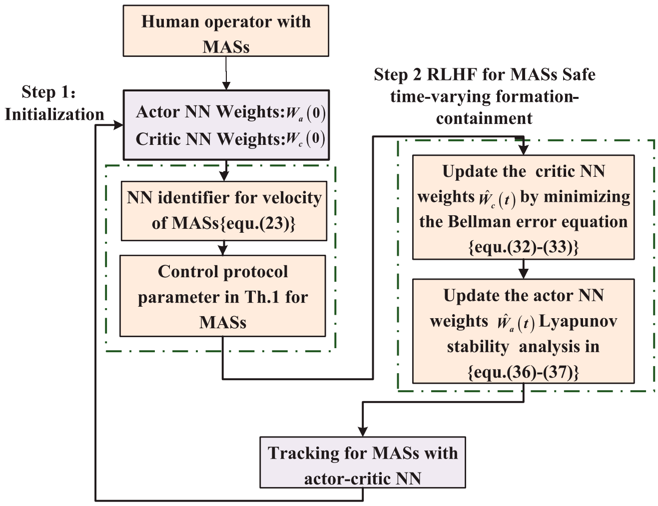

Fig. 3. Formation-containment tracking diagram.

TABLE I
INITIAL STATE OF MASs

UAV index i	(x(0),y(0))	vx(0)	vy(0)
0	(0,0) m	0 m/s	0 m/s
1	(0,0.5) m	1 m/s	1 m/s
2	(0,1) m	1 m/s	1 m/s
3	(0, –2) m	0 m/s	0 m/s
4	(0, –2.5) m	0 m/s	0 m/s
5	(0, –0.5) m	1 m/s	1 m/s
6	(0, –1) m	1 m/s	1 m/s
7	(0, –1.5) m	1 m/s	1 m/s
8	(–1,1) m	1 m/s	1 m/s
9	(–1, –2) m	0 m/s	0 m/s
10	(0, –2.5) m	0 m/s	0 m/s
11	(–1, –0.5) m	1 m/s	1 m/s
12	(–1, –1) m	1 m/s	1 m/s
13	(–1, –1.5) m	1 m/s	1 m/s
14	(–0.5, –0.5) m	1 m/s	1 m/s

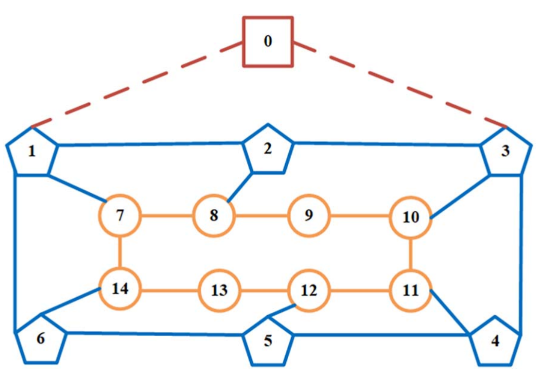

Fig. 4. MASs Communication topology.

$x = [x_1, x_2]^T$ denotes the center position of the ith UAV, $v = [v_1, v_2]^T$ indicates the velocity of ith UAV, respectively. The initial position and velocity of MASs are shown in Table I, respectively.

The reference signals of the TL can be described as

$$\dot{x}_0(t) = v_0(t), \dot{v}_0(t) = \begin{bmatrix} 2\cos(0.7t) + u_{\text{human1}} \\ u_{\text{human2}} \end{bmatrix} \tag{64}$$

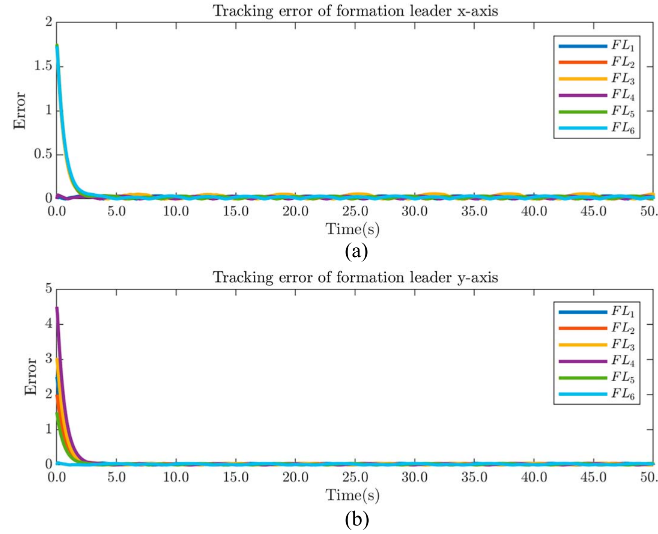

Fig. 5. Position error of formation leaders.

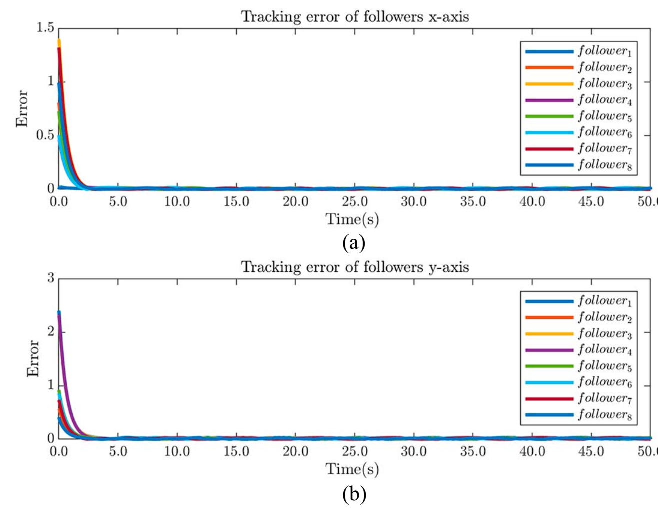

Fig. 6. Position error of followers.

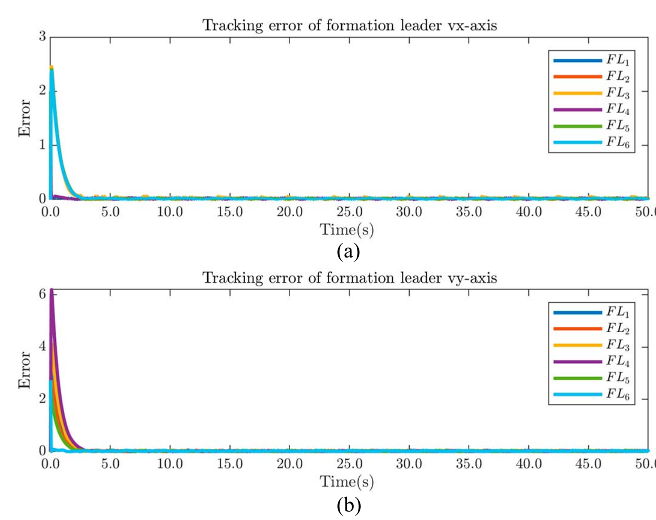

Fig. 7. Velocity error of formation leaders.

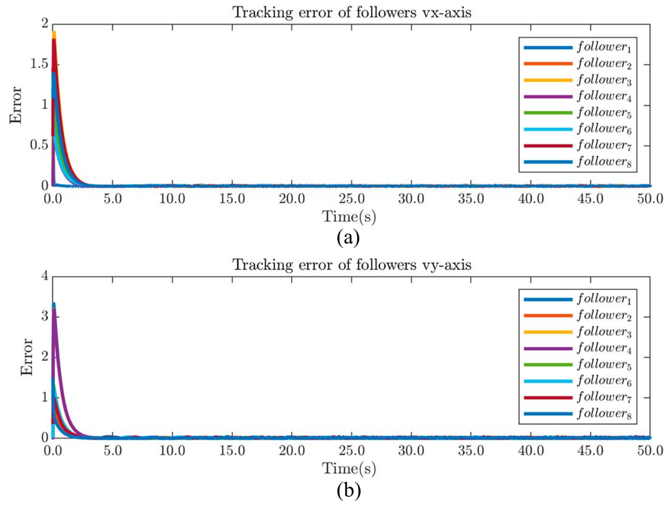

Fig. 8. Velocity error of followers.

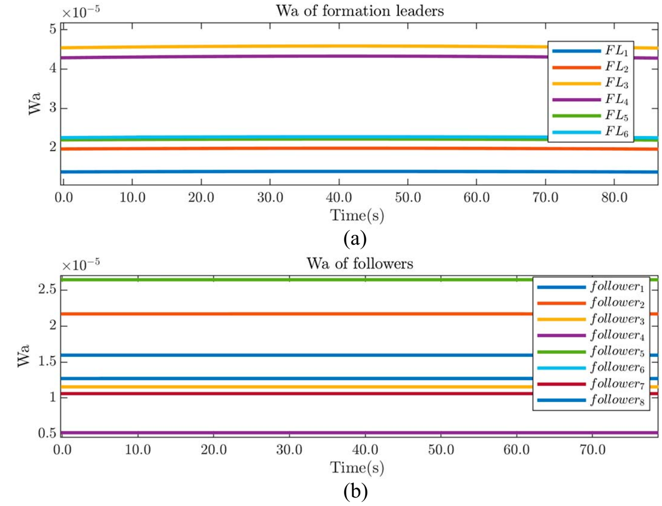

Fig. 10. Curves of actor weight for formation-leaders and containment followers.

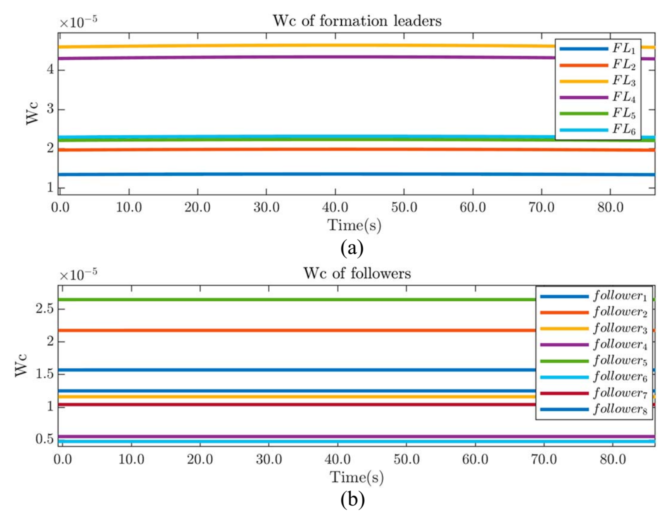

Fig. 9. Curves of critic weight for formation-leaders and containment followers.

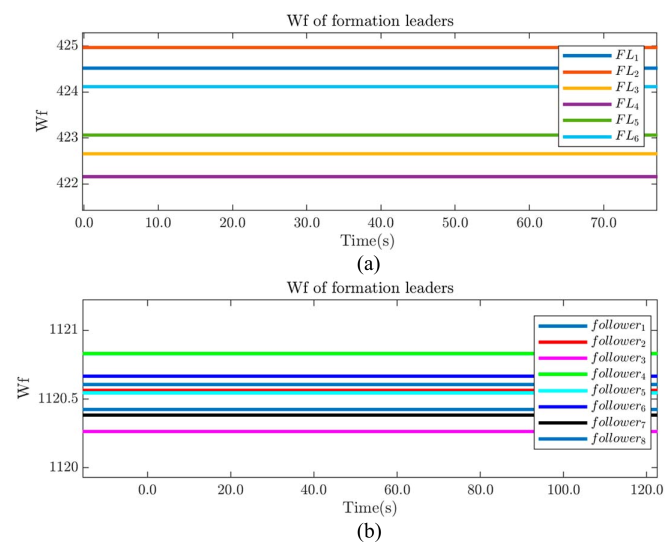

Fig. 11. Curves of identification weight. (a) Formation-leaders identification state errors. (b) Followers identification state errors.

Define $u_{\text{human}} = [u_{\text{human1}}, u_{\text{human2}}]^T$ is the human operator interaction nonautonomous tracking leader's control input to avoid hazard.

The initial position of the identifier $\hat{W}_{fi}(0) = [1, 1]^T$ $i \in 1, \ldots, 14$, which contains 101 nodes with centers μ_i evenly spaced in the range $[-6, 6]$ with the width. The connection topology among 15 UAVs is depicted in Fig. 4. Additionally, the designed parameters for the safe optimized formation containment $\tau_{Ex_i} = \tau_{Fx_i} = \text{diag}(60)_{2\times 2}$, $\tau_{Ev_i} = \tau_{Fv_i} = \text{diag}(40)_{2\times 2} i = 1 \ldots 14$ The nodes designed for actor–critic NNs is also 101, and the NN weight initial values are $\hat{W}_{eai}(0) = \hat{W}_{fai}(0) = \hat{W}_{eci}(0) = \hat{W}_{fci}(0) = [0.3]_{101\times 2}$.

Additionally, it is assumed that there exists an imperceptible obstacle in the environment during the movement. Thus, the human operator can interact with TL and communicate through the MASs network to avoid the hazard safely. Therefore, the control signal $u_0 = u_{\text{human}}$ of the TL can be considered as

$$u_0 = \begin{cases} \begin{bmatrix} -0.094 * \sin(\pi t(i)/40) + 0.1 \\ 0.062 * \cos(\pi t(i)/25) \end{bmatrix} & t \leq t_1 \\ \begin{bmatrix} -0.15 * \sin(\pi(t(i) - 30)/40) + 0.1 \\ -0.05 * \cos(\pi(t(i) - 30)/25) \end{bmatrix} & t_1 < t \leq t_2 \\ \begin{bmatrix} -0.02 * \sin(\pi(t(i) - 120)/40) + 0.1 \\ 0.15 * \cos(\pi(t(i) - 120)/25) \end{bmatrix} & t_2 < t \leq t_3. \end{cases} \tag{65}$$

where $t_1 = 10\,\text{s}, t_2 = 30\,\text{s}, t_3 = 50\,\text{s}$.

Figs. 5–8 show the formation leader and local containment tracking position and velocity error curves for each FL and the local containment error curve for each follower and all tracking errors converge to 0.

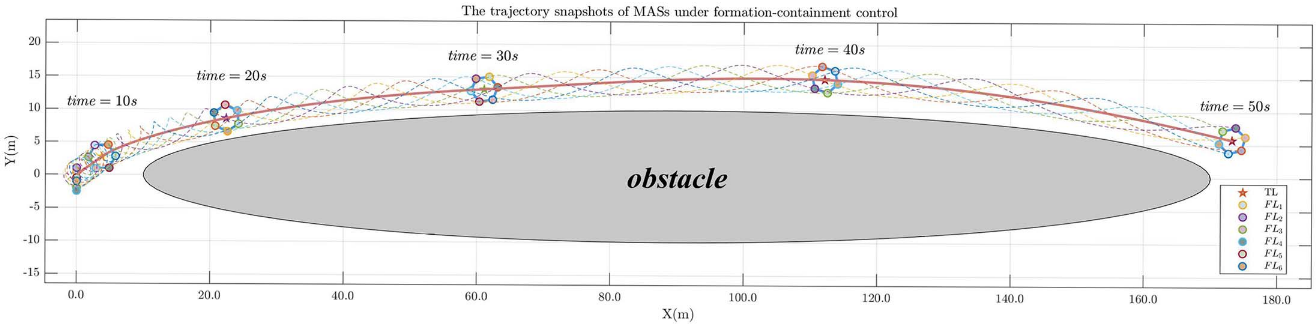

Fig. 12. Trajectory with HF.

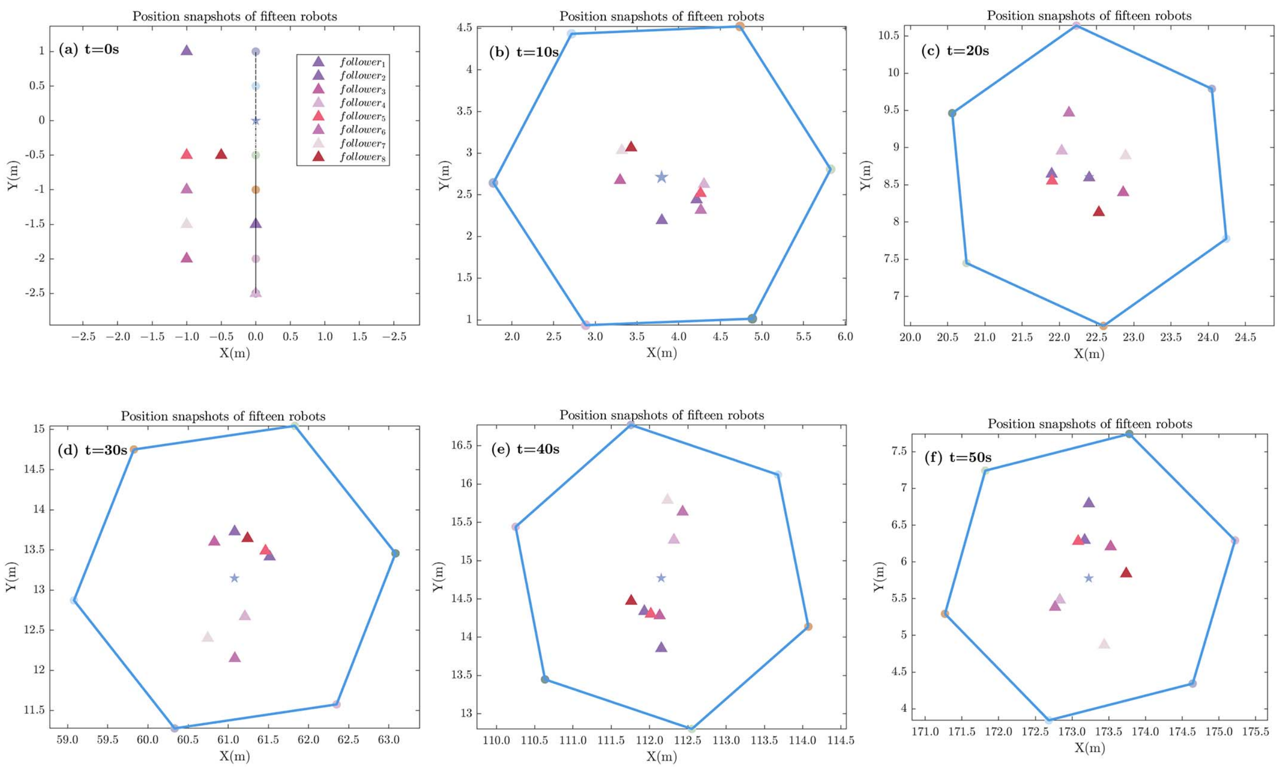

Fig. 13. Position snapshots of the fifteen robots.

Fig. 9 shows the curves of critic NNs weight in (32)–(33). Actor NNs weight in (36)–(37) is shown in Fig. 10. The NNs parameter of identified dynamics in (23) is shown in Fig. 11. It can be seen that all the NNs parameter can converge asymptotically. Fig. 12 demonstrates the process of completing formation containment tracking with obstacles, where the followers represented by the triangles zoomed in Fig. 13. The followers converge to the desired convex, which consists of FLs represented by the circles, and the FLs simultaneously track the given trajectory of TL represented by the five-pointed star.

Remark-3 (R.3): The HITLRL algorithm proposed in this article employs reinforcement learning integrated with online feedback technology, enabling the adjustment of control inputs based on current state information. This system is capable of rapidly responding to and correcting deviations in the face of fluctuations and disturbances, effectively reducing errors and perturbations. As a result, the simulation of the HITLRL algorithm can more closely simulate real-world scenarios. The algorithm is designed to be robust against potential delays in human input, errors in human input, sensor noise, communication latency, etc.

Remark-4 (R.4): With sufficient computational resources, the algorithm exhibits strong scalability, potentially accommodating a large-scale MAS.

V. Conclusion

In this article, the optimal formation containment tracking problems of second-order nonlinear MASs with unknown dynamics were investigated. First, the designed NNs parameter can identify the unknown nonlinear dynamics of FLs and followers and the identified error can converge asymptotically. Second, the actor–critic structure network inspired by the RL is designed to solve the nonlinear HJB equation for optimal formation containment tracking control, where tracking errors converge asymptotically to zero. Additionally, the human operator can interact with TL and communicate through the MASs network by using local topology information to achieve tacking safely in the obstacle environment. In the future, investigation into the utility of directed graphs and the practical application of heterogeneous nonlinear MASs on optimal time-varying formation-containment tracking control issues is essential.

References

[1] H. Zhao, H. Wang, N. Xu, and X. Zhao, "Fuzzy approximation-based optimal consensus control for nonlinear multi-agent systems via adaptive dynamic programming," *Neurcomputing*, vol. 553, 2023, Art. no. 126529.

[2] S. Jian, J. Zhang, L. Liu, Y. Wu, and Q. Shan, "Output consensus control of multi-agent systems with switching networks and incomplete leader measurement," *IEEE Trans. Auto. Sci. Eng.*, vol. 21, no. 4, pp. 6643–6652, Oct. 2024.

[3] W. Shi, G. Huang, S. Song, Z. Wang, T. Lin, and C. Wu, "Self-supervised discovering of interpretable features for reinforcement learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2712–2724, Nov. 2020.

[4] X. Wang, W. Guang, T. Huang, and J. Kurths, "Optimized adaptive finite-time consensus control for stochastic nonlinear multi-agent systems with non-affine nonlinear faults," *IEEE Trans. Auto. Sci. Eng.*, vol. 21, no. 4, pp. 5012–5023, Aug. 2023.

[5] D. Luo, Y. Wang, Z. Li, Y. Song, and F. Lewis, "Asymptotic leader-following consensus of heterogeneous multi-agent systems with unknown and time-varying control gains," *IEEE Trans. Auto. Sci. Eng.*, vol. 22, pp. 2768–2779, 2025.

[6] X. Gong, Y. Cui, J. Shen, J. Huang, Z. Wu, and W. Jiang, "Necessary and sufficient conditions of formation-containment control of high-order multiagent systems with observer-type protocols," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 7002–7016, Jul. 2022.

[7] L. Wei, M. Chen, and T. Li, "Disturbance-observer-based formation-containment control for UAVs via distributed adaptive event-triggered mechanisms," *J. Franklin. Inst.*, vol. 358, no. 10, pp. 5305–5333, 2021.

[8] L. Chen, S. Dai, and C. Dong, "Adaptive optimal tracking control of an underactuated surface vessel using actor–critic reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 16, no. 4, pp. 1–14, Nov. 2022.

[9] L. Ma, F. Zhu, and X. Zhao, "Human-in-the-loop formation-containment control for multi-agent systems: An observer-based distributed unknown input reconstruction method," *IEEE Trans. Control Netw. Syst.*, vol. 10, no. 4, pp. 2178–2189, Dec. 2023.

[10] W. Jiang, G. Wen, Z. Peng, T. Huang, and J. Cao, "Fully distributed formation-containment control of heterogeneous linear multiagent systems," *IEEE Trans. Autom. Control*, vol. 64, no. 9, pp. 3889–3896, Sep. 2019.

[11] B. Zhong, B. Bai, J. Li J, Y. Zhang, and Y. Fu, "Hierarchical tracking by reinforcement learning-based searching and coarse-to-fine verifying," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2331–2341, May 2019.

[12] L. Wang, D. Zhu, W. Pang, M. Wang, and C. Luo, "A novel obstacle avoidance consensus control for multi-AUV formation system," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 5, pp. 1304–1318, May 2023.

[13] X. Ge, Q.-L. Han, J. Wang, and X.-M. Zhang, "A scalable adaptive approach to multi-vehicle formation control with obstacle avoidance," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 6, pp. 990–1004, Jun. 2022.

[14] J. P. Cui, Y. N. Pan, H. Xue, and L. H. Tan, "Simplified optimized finite-time containment control for a class of multi-Agent systems with actuator faults," *Nonlinear Dyn.*, vol. 109, no. 7, pp. 2799–2816, Jun. 2022.

[15] J. Gonzlez-Sierra, M. Ramirez-Neria, J. Santiaguillo-Salinas, and E. G. Hernandez-Martinez, "Saturated formation containment control for a heterogeneous multi-agent system with unknown perturbations," *Automatica*, vol. 159, pp. 111343–111352, Oct. 2024.

[16] Z. W. Liu, H. Hou, and Y. W. Wang, "Formation-containment control of multiple under-actuated surface vessels with sampling communication via hierarchical sliding mode approach," *ISA Trans.*, vol. 124, pp. 458–467, 2022.

[17] X. L. Shao, H. C. Liu, W. D. Zhang, J. Zhao, and Q. Z. Zhang, "Path driven formation-containment control of multiple UAVs: A path-following framework," *Aerosp. Sci. Technol.*, vol. 135, pp. 108168–108187, Apr. 2023.

[18] Q. Liu, H. Yan, M, and Wang, M., "Data-driven optimal bipartite consensus control for second-order multi-agent systems via policy gradient reinforcement learning," *IEEE Trans. Cybern.*, vol. 54, no. 6, pp. 3468–3478, Jun. 2024.

[19] J. Wu, Z. Huang, Z. Hu, and C. Lv, "Toward human-in-the-loop AI: Enhancing deep reinforcement learning via real-time human guidance for autonomous driving," *Engineering*, vol. 21, pp. 75–91, 2023.

[20] Z. Qin, H. N. Wu, and J. L. Wang, "Behavior learning based distributed tracking control for human-in-the-loop multi-agent systems," *J. Franklin Inst.*, vol. 360, no. 13, pp. 9996–10017, 2023.

[21] M. Li, F. L. Zhu, and X. D. Zhu, "Human-in-the-loop consensus control for multi-agent systems with external disturbances," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 11024–11034, Aug. 2024.

[22] Y. Han et al., "Deep reinforcement learning for robot collision avoidance with self-state-attention and sensor fusion," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 6886–6893, Jul. 2022.

[23] G. Wen, S. S. Ge, and F. Tu, "Optimized back-stepping for tracking control of strict-feedback systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3850–3862, Aug. 2018.

[24] X. Zhao, S. Chen, Z. Zhang, and Y. Zheng, "Consensus tracking for high-order uncertain nonlinear MASs via adaptive back-stepping approach," *IEEE Trans. Cybern.*, vol. 53, no. 2, pp. 1248–1259, Feb. 2023.

[25] K. S. Hwang, S. W. Tan, and M. C. Tsai, "Reinforcement learning to adaptive control of nonlinear systems," *IEEE Trans. Syst., Man, Cybern., B, Cybern.*, vol. 33, no. 3, pp. 514–521, Jun. 2003.

[26] H. Xu, S. Li, D. Yu, C. P. Chen, and T. Li, "Adaptive swarm control for high-order self-organized system with unknown heterogeneous nonlinear dynamics and unmeasured states," *Neurocomputing*, vol. 440, pp. 24–35, Jun. 2021.

[27] Y. Qian, Z. Miao, J. Zhou, and X. Zhu, "On consensus control of nonlinear multi-agent systems satisfying incremental quadratic constraints," *J. Franklin Inst.*, vol. 360, no. 12, pp. 9240–9264, 2023.

[28] Y. Jiang, J. Lv, C. Wang, Y. Kao, and F. Wang, "Prescribed-time disturbance observer based fully distributed prescribed-time containment control of multi-agent systems," *IEEE Trans. Circuit Syst. II: Exp. Briefs*, vol. 71, no. 4, pp. 2044–2048, Apr. 2024.

[29] H. Zhou and S. Tong, "Adaptive neural network event-triggered output-feedback containment control for nonlinear MASs with input quantization,", *IEEE Trans. Cybern.*, vol. 53, no.11, pp. 7406–7416, 2023.

[30] G. Xie, H. Xu, Y. Li, C. D. Wang, B. Zhong, and X. Hu, "Consensus seeking in large-scale multi-agent systems with hierarchical switching-backbone topology," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15810–15824, Nov. 2024.

[31] W. Yun, S. Park, J. Kim, M. Shin, S. Jung, and D. Mohaisen, "Cooperative multiagent deep reinforcement learning for reliable surveillance via autonomous multi-UAV control," *IEEE Trans. Ind. Inform.*, vol. 18, no. 10, pp. 7086–7096, Oct. 2022.

[32] M. Ran, J. Li, and L. Xie, "Reinforcement-learning-based disturbance rejection control for uncertain nonlinear systems," *IEEE Trans. Cybern.*, vol. 52, no. 9, pp. 9621–9633, Sep. 2022.

[33] L. Chen, Y. Guo, G. Ma, Y. Li, and B. Xiao, "Formation-containment control of multi-agent systems with communication delays," *ISA Trans.*, vol. 128, pp. 32–43, 2022.

[34] D. Mei, J. Sun, Y. Xu, and L. Dou, "Model-free learning-based distributed cooperative tracking control of human-in-the-loop multi-agent systems," *Int. J. Robust Nonlinear Control*, vol. 34, no. 11, pp. 7333–7349, Mar. 2024.

[35] T. Wang, X. Dong, Y. Hua, D. Wang, and Z. Ren, "Practical optimal formation-containment tracking control of nonlinear multi-agent systems with unknown dynamics," *IEEE Syst. J.*, vol. 17, no. 4, 6564–6575, Dec. 2023.

[36] C. Wei, X. Wu, B. Xiao, J. Wu, and C. Zhang, "Adaptive leader-following performance guaranteed formation control for multiple spacecraft with collision avoidance and connectivity assurance," *Aerosp. Sci. Technol.*, vol. 120, pp. 107266–107277, Jan. 2022.

[37] Z. Du, H. Wang, and H. Yan, "Model predictive formation tracking-containment control for multi-UAVs with obstacle avoidance," *IEEE Trans. Syst. Man Cybern. Syst.*, vol. 54, no. 6, pp. 3404–3414, Jun. 2024.

[38] Q. Wei, X. Wang, X. Zhong, and N. Wu, "Consensus control of leader-following multi-agent systems in directed topology with heterogeneous disturbances," *IEEE/CAA J. Automat. Sin.*, vol. 8, no. 2, pp. 423–431, Feb. 2021.

[39] G. S. Seyboth, D. V. Dimarogonas, and K. H. Johansson, "Event-based broadcasting for multi-agent average consensus," *Automatica*, vol. 49, no. 1, pp. 245–252, Jan. 2013.

Luning Yang is currently working toward the Ph.D. degree in artificial intelligence with the Institute of Artificial Intelligence, Beihang University, Beijing, China.

Her research interests include multi-agent systems, reinforcement learning, optimal control, and path planning.

Pei Chi received the B.S. and Ph.D. degrees in navigation, guidance, and control from Beihang University, Beijing, China, in 2002 and 2008, respectively.

Since 2020, he has been a Professor with Beihang University, where he is currently the Vice President of the Unmanned Systems Research Institute. He was a Senior Engineer and a Researcher with the headquarters of Aviation Industry Corporation of China, from 2008 to 2020. His research interests include multi-agent reinforcement learning, swarm intelligence, and flight path planning.

Jiang Zhao is currently an Associate Professor in the field of navigation, guidance, and control with the School of Automation Science and Electrical Engineering, Beihang University, Beijing, China. He was selected into the "Top Ten Scientific Problems" team of the cross integration of science and technology and the "Young Top Talents" program at Beihang University and into the "Annual Scientific Influence" list of the world's top 2%; top scientists in the aerospace field in 2023. He is selected as an excellent reviewer of *ISA Transactions*, *Acta Astronautica*, and *Aerospace Science and Technology*. His research interests include multi-agent systems, reinforcement learning, and path planning.

Yingxun Wang is currently a Professor in the field of navigation, guidance, and control with the Institute of Unmanned Systems, Beihang University, Beijing, China.

Dr. Wang is a member of the Chinese Society of Aeronautics and Astronautics, the Honorary Director of GNC Branch, the Deputy Director of the Unmanned Aircraft Special Committee, the Director of UAV Expert Committee of China Aircraft Owners and Pilots Association, and the Editor-in-Chief of *Guidance, Navigation and Control* international journal. His research interests include multi-agent systems, reinforcement learning, autonomous trajectory optimization, and path planning.